

Selecting Assembled PCB Desktop MANUFACTURING EQUIPMENT



Nidhi Kathuria
is a senior
application
engineer at EFY
Labs, New Delhi

This buyers' guide explains how desktop manufacturing equipment can be used for full product realisation and to understand how production considerations inevitably impact the design process. When designing a product the most important factor is defining a product's success. However, with advancements in desktop manufacturing, most parts can now be produced in-house, which saves a lot of time.

Presented in this article are the different technologies and buying tips for the most popular types of desktop manufacturing equipment. For a new product, the first thing is the product realisation process, which encompasses the entire cycle of production and mechanical design so that all components including assembled printed circuit boards (PCBs), are designed to fit into the mechanical design.

Then, if everything is okay, PCB designs are verified and finalised and their Gerber files are sent for production of the PCBs to the PCB manufacturer. Sometimes, manufacturers may revise these files before they proceed further. Also, they may provide turn-key solutions, where they make PCBs, source components, make stencils and assemble PCBs, all at one place to save time, in order to meet the pace of the market.

Key factors involved in PCB manufacture

Basic equipment for PCB production on the basis of the board as input are the screen

or stencil printer, pick-and-place machine, reflow oven, board DE-fluxing machine, stencil cleaning machine and automated optical inspection system. Steps involved in producing a PCB are:

1. The screen printer applies solder paste to the PCB on pad locations.
2. The pick-and-place machine picks up, inspects and puts surface-mount technology (SMT) components in a particular position on programmed locations.
3. Board DE-fluxing and stencil cleaning machines are used for cleaning the PCB and stencil, respectively.
4. Automated inspection equipment are used for inspection of the assembled PCBs for detection of any wrong placement or missing components.

Electronics manufacturing using SMT is a method in which electronic components are assembled with automated machines that place components on the surface of a board. The process involves making the PCB and assembling the components on that PCB.

After checking availability of various types of desktop manufacturing equipment, you should consider the various types available and be familiar with the associated pros and cons. All PCB prototyping work can be done in-house and time-to-market can be reduced.

Selection criteria

Stencil printers. The first stage of automated assembling of PCBs is stencil printing. For this we require stencil, solder paste and printer. Components are either leaded or SMT soldered to the board and exposed areas of the board are tinned or plated with solder. Stencil printers are available in three configurations, as given below:

Manual. Simple printing devices are suitable where high-precision alignment of a stencil-PCB is not critical. These are

Fig. 1: Desktop stencil printer

